

Review

Effects of antipredator training, environmental enrichment, and soft release on wildlife translocations: A review and meta-analysis

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ABSTRACT

Wildlife translocations can have conservation value but results have been mixed regarding animal behavior and survival post-release. Practitioners have adopted antipredator training, environmental enrichment, and soft release as pre-release conditioning tactics to encourage adaptive behavior and improve post-release survival, but their utility has not been broadly quantified. We performed a formal literature review and conducted meta-analysis on 108 effects from 41 studies experimentally testing how these tactics affected survival, movement, or site fidelity compared to unconditioned animals. We further investigated how each conditioning tactic, animal source (wild-to-wild translocated or captive-released), age, and taxonomic group (birds, fish, mammals, and reptiles) influenced outcomes. Relative to unconditioned animals, conditioned individuals were 1.5 times more likely to survive, had reduced movement, and were three times more likely to show site fidelity. Each of the three conditioning tactics resulted in improved survival. Juveniles released from captivity derived the greatest survival benefit from conditioning. Across taxa, conditioning most benefitted survival of fish. Conditioning also had positive effects on survival of mammals and reptiles, albeit with less certainty than for fish. Estimates comparing survival of conditioned to unconditioned birds were much more variable, suggesting avian translocation programs using conditioning generally need improvement. Soft release consistently reduced movement and increased site fidelity; this was an especially viable technique for adult wild-to-wild translocated animals. We provide quantitative evidence that behavioral conditioning can aid wildlife translocations, and we encourage continued experiments to further elucidate how refined tactics could advance conservation efforts using translocation as a management tool.

1. Introduction

Conservation translocations entail deliberate movement of organisms from one site for release in another, often with the aim of augmenting imperiled populations or reintroducing species to areas where they have been extirpated (IUCN/SSC, 2013). Conservation translocations (hereafter “translocations”) have been used for decades as a wildlife management tactic, but success of such efforts has generally been low or uncertain (Griffith et al., 1989; Fischer and Lindenmayer, 2000; Seddon et al., 2007). Failure of released populations to establish and persist could result from animals exhibiting behaviors associated with low post-release survival such as maladaptive habitat selection and atypical movement (Einum and Fleming, 2001; Jule et al., 2008; Swaisgood, 2010). The effectiveness of translocation is still critically questioned (Pérez et al., 2012; Taylor et al., 2017), and the utility of long-standing tactics thought to contribute to success needs to be

quantified to advance the field of reintroduction biology (Armstrong and Seddon, 2008). Tactics aimed at conditioning the behavior of translocated animals, such as environmental enrichment, antipredator training, and soft release are rapidly gaining popularity as means to improve translocation success (Reading et al., 2013; Batson et al., 2015; Shier, 2016), but their utility has not been broadly quantified. We evaluated the effectiveness of behavioral conditioning on translocations releasing captive and wild animals to examine which factors related to these approaches influence success.

Captive animals are an important source for translocations (Kleiman, 1989; Brichieri-Colombi et al., 2018), but captivity can have detrimental effects on behavior (DeGregorio et al., 2017; Swaisgood et al., 2018). Offsetting these effects is often accomplished using environmental enrichment. This can take on numerous forms but generally entails providing environmental stimuli necessary for optimal psychological and physiological wellbeing (Shepherdson, 1998).

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Enrichment can enhance translocations by providing essential learning opportunities pre-release (Reading et al., 2013). For example, rearing black-footed ferrets (*Mustela nigripes*) in outdoor pens and giving them experience killing live prey was shown to reduce dispersal and improve survival compared to ferrets reared in simplistic cages (Biggins et al., 1999). Enrichment also might help captive animals avoid developing maladaptive behaviors. For instance, practitioners captive-rearing birds for translocation frequently use puppets, costumes, or parent-rearing to reduce the likelihood of juveniles imprinting on human caretakers (Valutis and Marzluff, 1999; Utt et al., 2008). Likewise, stranded southern sea otter (*Enhydra lutris nereis*) pups had better foraging skills and increased survival when reared with a surrogate adult female than if rehabilitated using traditional methods relying heavily on human care (Nicholson et al., 2007). Yet, it is unclear how broadly effective environmental enrichment is for enhancing translocation success, as it had minimal to no effects on post-release behavior and survival for other mammals (Rogers et al., 2016) as well as for some fish (Carrera-García et al., 2017) and reptiles (Roe et al., 2015; DeGregorio et al., 2017).

Although environmental enrichment provides opportunities for captive animals to express several natural behaviors, it might be ineffective at directly teaching individuals to be wary of predators (Griffin et al., 2000). Naïve animals often do not behaviorally demonstrate recognition of relevant predators (Blumstein et al., 2002), which is concerning because predation is a prevailing cause of translocation failure (Fischer and Lindenmayer, 2000). Antipredator training using live or model predators as well as predator cues (e.g., scent) can be an effective approach for conditioning aversive behavior towards predators. For example, juvenile black-tailed prairie dogs (*Cynomys ludovicianus*) exposed to live predators before translocation had higher post-release survival than untrained conspecifics (Shier and Owings, 2006), and Chinook salmon (*Oncorhynchus tshawytscha*) had greater post-release recovery rates if fish were exposed to live predators prior to release (Maynard et al., 1998). However, antipredator training does not always improve post-release survival. For instance, Lopes et al. (2017) used model predators and human contact paired with aversive stimuli to condition blue-fronted Amazon parrots (*Amazona aestiva*), and Moseby et al. (2012) used model, tactile, and olfactory predator stimuli to train greater bilbies (*Macrotis lagotis*); neither study found apparent benefits of conditioning.

Whether animals are released from captivity or free-ranging individuals are translocated between natural sites, challenges incurred with moving them to novel environments such as stress (Teixeira et al., 2007) and lack of familiar sensory cues (Swaigood, 2010) could be associated with failure. As such, animals often rapidly disperse from release areas, which could limit reproductive potential and reduce survival by unnecessarily expending energy and increasing exposure to predators or vehicles (Le Gouar et al., 2012). To combat such effects, practitioners have employed “delayed” or “soft” release measures to aid animals’ transition to release sites. This typically involves temporarily confining animals in pens, providing supplementary resources (e.g., food, water, access to shelter), or implementing a combination of penning and resource provisioning at the release site (Cid et al., 2014). The rationale for soft release is that animals have greater chance to acclimate and become anchored to their new surroundings and could be less apt to immediately disperse once released. However, results from studies using soft release have also been mixed. Compared to animals that are immediately released unrestrained into the recipient site and not provisioned with supplementary resources (i.e., hard released), some authors reported soft release reduced dispersal, increased site fidelity, and improved survival (Tuberville et al., 2005; Mitchell et al., 2011; Knox and Monks, 2014). Yet, others reported minimal to no effects (Hardman and Moro, 2006; de Milliano et al., 2016; Bannister et al., 2018) or even suggest it can be a detriment to success (Thompson et al., 2001; Richardson et al., 2013; Batson et al., 2017).

Clearly there is considerable discrepancy regarding the efficacy of

conditioning on behavior and survival of translocated animals, perhaps because the basis for using these tactics often comes from intuition rather than evidence (Seddon et al., 2007). Quantitatively evaluating the effectiveness of pre-release conditioning will help identify successful tactics from those needing refinement and provide valuable insight to inform future translocations. To this end, we reviewed published translocation literature, and used meta-analysis of data from studies experimentally testing the effects of environmental enrichment, anti-predator training, and soft release to ask: Does conditioning broadly benefit translocations, and if so, is success affected by factors such as conditioning tactic, translocation type (wild-to-wild or captive releases), age, or taxonomic group? Given the variable nature of translocations and conditioning tactics used (Batson et al., 2015), we expected effect sizes generated from meta-analysis would be generally small. Based on benchmarks commonly used to evaluate short-term success, we predicted conditioned animals would have higher post-release survival, move less, and show greater fidelity to release areas than unconditioned animals.

2. Methods

2.1. Data sources, inclusion criteria, and extraction

We searched for literature current to May 2018 in Google Scholar using the following terms singly and in combination: “antipredator training”, “captive-”, “condition”, “conservation”, “delayed release”, “enrich-”, “enrichment”, “environmental”, “head-start”, “headstart”, “reinforce-”, “reintroduc-”, “soft release”, “tactic”, “translocat-”, “translocation”, and “wildlife”. This search returned peer-reviewed journal articles, technical reports, book chapters, and graduate theses and dissertations. We scanned titles and abstracts to determine if articles might be eligible for inclusion after further reading and applying our criteria (see below). To gather other potential sources for our dataset, we reviewed reference lists in (backward search) and citations of (forward search) articles meeting our criteria as well as relevant reviews (Reading et al., 2013; Batson et al., 2015). Although our search likely missed some applicable studies, we are confident the resulting dataset from our systematic search is a robust representation of the available literature (Nakagawa et al., 2017).

Many translocations are not conducted as experiments, precluding the ability to challenge historic assumptions about what interventions are beneficial and limiting mechanistic understanding of outcomes (Seddon et al., 2007; Swaigood, 2010; Bannister et al., 2018). Thus, to be included in meta-analysis, a study had to experimentally compare survival, movement, or site fidelity between conditioned animals and unconditioned conspecifics. We categorized conditioning tactics based on distinctions described in previous reviews (Reading et al., 2013; Batson et al., 2015). We included studies that used only environmental enrichment, antipredator training, or soft release as these are the most regularly tested tactics. However, because each of these tactics can entail a wide range of methods we restricted the studies used in our meta-analysis by narrowly defining the tactics as follows: We defined environmental enrichment as providing animals stimulation in captivity that gave them opportunities to express species-typical behaviors that are expected to confer post-release benefits. For example, housing animals in naturalistic enclosures, giving opportunities to forage like they would in nature, and interaction with conspecifics all constituted enrichment. We considered antipredator training separate from enrichment since this usually entails brief, infrequent sessions (van Heezik et al., 1999; Alonso et al., 2011; Cortez et al., 2015) whereas enrichment is often passively provided. We thus defined antipredator training as situations where researchers actively attempted to condition this specific behavior by simulating predators or providing exposure to live predators/predator sensory cues. We defined soft release by when animals were temporarily confined at the release site in some form of enclosure rather than being immediately released. We did not include

studies that used other soft release measures such as resource supplementation because this is rarely experimentally implemented (e.g., a treatment group with provisioning compared to an unprovisioned group). We were necessarily restrictive in our data inclusion to protect the integrity of the analyses, and as a result we excluded several studies that nearly, but not entirely, met our inclusion criteria. Forty articles were retained for data extraction after applying our inclusion criteria (Supplementary data). We also included data from an unpublished study on translocation of eastern box turtles (*Terrapene carolina*) conducted by the authors in Michigan, USA.

We created a separate dataset for each of our main effects of interest (survival, movement, and site fidelity) and analyzed these datasets independently. In several cases, a study investigated more than one of these effects and was thus used in multiple databases. Sample sizes of monitored released animals in each group (conditioned and unconditioned) had to be reported for inclusion so we could calculate effect sizes for analyses. Data were generally provided from animals monitored with telemetry, but if effects were estimated from other methods (e.g., recapture rates as a survival proxy), we converted proportions to whole numbers based on release sample sizes (Barron et al., 2010). To be included in the survival dataset, studies had to report the number or proportion of animals known or estimated to have survived and died during the post-release monitoring period. To be included in the movement dataset, studies had to report mean and standard deviation (SD) of metrics used to evaluate movement (e.g., distance dispersed from release sites, total area used, or daily movement rate). If SD was not reported but standard errors or symmetric confidence intervals were, we converted them to SD. To be included in the site fidelity database, authors had to report the number or proportion of animals that did or did not exhibit site fidelity according to their definition of what constituted fidelity. For instance, authors have defined site faithfulness as staying within 1 km of the release area (Tuberville et al., 2005; Hardman and Moro, 2006). For 10 studies where data were not readily extractable but met our inclusion criteria, we contacted authors up to two times for clarification; authors of four articles responded to our requests.

We made judgments of whether to retain effects from a given study on a case-by-case basis. If a study simultaneously conditioned a group with multiple tactics, we did not include effects from this treatment. For example, D'Anna et al. (2012) had three treatment groups for conditioning white seabream (*Diplodus sargus*): environmental enrichment, antipredator training, or both enrichment and predator training. Disentangling the effects of simultaneous conditioning from enrichment and antipredator training would not be possible, so we did not include effects from this group. If a study had replication such as multiple release cohorts or sites, we treated them as separate effects if authors reported this level of information. For example, Jonsson et al. (1999) soft released brown trout (*Salmo trutta*) in multiple areas, so we included effects from each site. Similarly, Maynard et al. (1996) reared Chinook salmon with either sand or gravel substrate. We considered both of these groups environmentally enriched but treated them as independent effects since the authors reported results for each group.

In addition to the abovementioned data, we assigned each study a unique ID and noted citation information such as authors, journal title (if a peer-reviewed article), and publication year; continent where the research was conducted; how long animals were monitored post-release; Latin and common name of the focal species; the main conditioning tactic as environmental enrichment, antipredator training, or soft release; whether translocations were wild-to-wild (free-ranging animals moved directly between natural sites) or captive releases. In nearly all cases, animals released from captivity were captive-born, but some wild-born animals were held and conditioned in captivity prior to translocation (e.g., Nicholson et al., 2007; DeGregorio et al., 2017). Only four vertebrate classes were present in our datasets, so we noted Class as Actinopterygii, Aves, Mammalia, or Reptilia. We recorded age as adult, juvenile, both if multiple age classes were translocated, or

unknown/not stated. We also noted sex as male, female, both if multiple sexes were translocated, or unknown/not stated.

2.2. Effect size calculation and meta-analyses

We calculated effect sizes and conducted all analyses in R version 3.4 (R Core Team, 2017). We used the *metafor* package (Viechtbauer, 2010) to calculate effect sizes and associated variances from extracted data using the *escalc* function and to construct meta-analytical models. We generated effect sizes for survival and site fidelity as odds ratios (OR) using the numbers of conditioned and unconditioned animals known or estimated to have survived and died and that did or did not exhibit site fidelity, respectively. Since we combined multiple metrics for our movement analysis, we calculated standardized mean difference of movement between conditioned and unconditioned animals as Hedge's *g* (also commonly referred to as Hedge's *d*) from means and SD. We considered effect sizes of 0.2, 0.5, and 0.8 as small, medium, and large, respectively, for Hedge's *g* (Nakagawa and Cuthill, 2007).

Controlling for phylogenetic non-independence is important when conducting meta-analyses with multiple species (Nakagawa et al., 2017). We created a phylogenetic tree of all species used in analyses with the *rotl* package (Michonneau et al., 2016). This package does not calculate tree branch lengths, so we estimated them using the *compute.brlen* function in the *ape* package (Paradis and Schliep, 2018). We then created phylogenetic correlation matrices from the resulting trees (Figs. A1–A3) if needed for use in models.

To determine if the random factors “phylogeny” and “study ID” were needed to control for phylogenetic non-independence or multiple effects analyzed from the same study, we used likelihood ratio tests (LRT) comparing models including these random factors to a null model with no random factors. Accepting statistical significance at $\alpha \leq 0.05$, the LRTs suggested both factors were needed for survival ($P \leq 0.04$) and movement ($P \leq 0.002$), so we used multilevel models to conduct meta-analyses of these effects. The LRTs suggested neither random factors were needed for site fidelity ($P \geq 0.60$), so we used a random effects meta-analytical model for this analysis.

Beyond estimating the overall magnitude of an effect, meta-analysis can be used to quantify the level of consistency among effect sizes used in analysis (Senior et al., 2016). We determined how much heterogeneity (inconsistency) existed among effects in each dataset using the I^2 statistic, which ranges from 0 to 100% (Nakagawa et al., 2017). Values of 25%, 50%, and 75% broadly suggest low, moderate, and high amounts of heterogeneity, respectively (Higgins et al., 2003). I^2 values $> 80\%$ are common in ecological and evolutionary meta-analyses since they often deal with multiple species and systems (Senior et al., 2016). We thus a priori anticipated there would be moderate to high levels of heterogeneity among effect sizes in our coarse meta-analyses, and this often suggests more detailed subsequent analyses will be useful for deciphering potential causes of variation among effect sizes (Senior et al., 2016; Nakagawa et al., 2017).

In addition to the overall effects of behavioral conditioning on survival, movement, and site fidelity, we were also interested in determining what factors might influence these metrics. We thus conducted several independent meta-analyses on subsets of effects in each of our datasets. Using the same model structures as in our main meta-analyses, we conducted subgroup analyses to investigate if survival, movement, or site fidelity were affected by conditioning tactic, translocation type, age, or Class. We conducted subgroup analyses only if there were at least five effects comprised of effects from three independent studies per subgroup within each dataset. Thus, we did not conduct subgroup analyses for certain categorical variables due to it not being tested, lack of reported information, or low numbers of effects within each category. Our subgroups for survival were conditioning tactic (environmental enrichment, antipredator training, or soft release), translocation type (wild-to-wild or released from captivity), age (adult or juvenile), and Class (Actinopterygii, Aves, Mammalia, or

Reptilia). Subgroups for movement were conditioning tactic (environmental enrichment and soft release only), translocation type, age, and Class (Mammalia and Reptilia only). Wild-to-wild translocations within translocation type were the only factor where enough effects existed in the site fidelity dataset to conduct subgroup analysis.

We report effect sizes with 95% confidence intervals (CI) for all meta-analytical models. We considered effects to be statistically significant if CIs for odds ratios of survival and site fidelity did not overlap one. For movement, we considered effects to be statistically significant if CIs did not overlap zero for Hedge's *g* estimates. However, we also interpret results in terms of biological importance even if some estimates had CIs overlapping traditional significance threshold criteria (Nakagawa et al., 2017).

Publication bias can arise when studies with non-significant results are less likely to be published. We investigated for the potential of publication bias in each dataset by inspecting funnel plots for asymmetry, running Egger's regression tests (Egger et al., 1997), and conducting trim-and-fill analyses to determine the number effects that might have been "missing" from our datasets due to publication bias (Duval and Tweedie, 2000). We included effects only from peer-reviewed studies in these analyses.

3. Results

3.1. Summary of all effects

In total, we analyzed 108 effects from 41 studies meeting our inclusion criteria: 59 effects from 36 studies of survival, 41 effects from 18 studies of movement, and eight effects from six studies of site fidelity. There was a range of 1–5 effects per study for survival, 1–6 effects per study for movement, and 1–2 effects per study for site fidelity. The most commonly used conditioning tactic was soft release. We analyzed 73 effects for soft release, 28 effects for environmental enrichment, and seven effects for antipredator training. There were 70 effects for captive releases and 38 for wild-to-wild translocations. Fifty effects were for adults, 49 effects were for juveniles, and nine effects were from multiple age groups. There were 36 effects for reptiles, 33 for mammals, 22 for fish, and 17 for birds. Most studies released both males and females (72 effects), only one study released solely males (one effect) or females (three effects), but many studies did not report or were uncertain of sexes (32 effects). Forty-six effects were from studies conducted in North America, 29 from Australia/New Zealand, 25 from Europe, seven from Asia, and only one from South America. Eighty-seven effects were from peer-reviewed articles in 23 different journals, 16 were from gray literature (three from books, two from technical reports, and 11 from graduate theses), and five were from an unpublished dataset. Publication dates spanned 37 years (1981–2018).

3.2. Survival

Soft release had the most effects for survival, followed by environmental enrichment, and antipredator training had the least (Table 1). There were more effects on captive releases than wild-to-wild translocations. There were slightly more effects for juveniles than adults, and only a few effects were from multiple age classes. The largest number of effects was for mammals, followed by fish, birds, and reptiles.

Overall, translocated animals were 1.55 (95% CI: 1.23 to 1.95) times more likely to survive post-release if they had been conditioned (Fig. 1). As expected, we detected moderate heterogeneity between effects ($I^2 = 42.16\%$). The application of any of the three conditioning tactics resulted in higher survival relative to unconditioned animals: Animals receiving antipredator training were 2.14 (95% CI: 1.34 to 3.40) times more likely to survive than untrained ones, environmentally enriched animals were 1.55 (95% CI: 1.01 to 2.39) times more likely to survive than unenriched individuals, and soft released animals were 1.47 (95% CI: 1.07 to 2.02) times more likely to survive than hard

Table 1

Contingency tables showing the distribution of effects for survival of conditioned and unconditioned translocated animals based on age class, taxonomic group, conditioning tactic, and translocation type. NA refers to nonsensical effects (e.g., wild-to-wild translocations would not use environmental enrichment since this conditioning occurs in captivity).

Age class	Taxa			
	Actinopterygii	Aves	Mammalia	Reptilia
Adult	0	6	15	2
Juvenile	18	7	3	4
Multiple	0	2	2	0

Conditioning tactic	Age class		
	Adult	Juvenile	Multiple
Antipredator training	2	4	0
Soft release	19	15	4
Environmental enrichment	2	13	0

Conditioning tactic	Taxa			
	Actinopterygii	Aves	Mammalia	Reptilia
Antipredator training	1	4	1	0
Soft release	8	10	17	3
Environmental enrichment	9	1	2	3

Conditioning tactic	Translocation type	
	Captive release	Wild-to-wild
Antipredator training	5	1
Soft release	23	15
Environmental enrichment	15	NA

Translocation type	Age class		
	Adult	Juvenile	Multiple
Captive release	10	32	1
Wild-to-wild	13	0	3

Translocation type	Taxa			
	Actinopterygii	Aves	Mammalia	Reptilia
Captive release	18	10	10	5
Wild-to-wild	0	5	10	1

released animals. Animals released from captivity were 1.73 (95% CI: 1.25 to 2.40) times more likely to survive if they had been conditioned, but we found minimal evidence suggesting conditioning benefited survival for wild-to-wild translocations (OR = 1.09, 95% CI: 0.64 to 1.87). Juveniles were 1.94 (95% CI: 1.17 to 3.21) times more likely to survive if they had been conditioned, but conditioning had more variable effects for adults (OR = 1.31, 95% CI: 0.82 to 2.09). Conditioning led to improved survival for fish (OR = 1.51, 95% CI: 1.13 to 2.03). Conditioning also had generally positive benefits for mammals (OR = 1.38, 95% CI: 0.85 to 2.24) and reptiles (OR = 1.95, 95% CI: 0.78 to 4.88), but not with the precision it did for fish. Although the mean effect size was similar to other taxa, there was much more uncertainty in the effectiveness of conditioning for birds (OR = 1.59, 95% CI: 0.25 to 9.95). We found no evidence of publication bias from the funnel plot or the Egger's regression test ($z = 0.33$, $P = 0.74$). The trim-and-fill analysis estimated no effects were missing (Fig. A4).

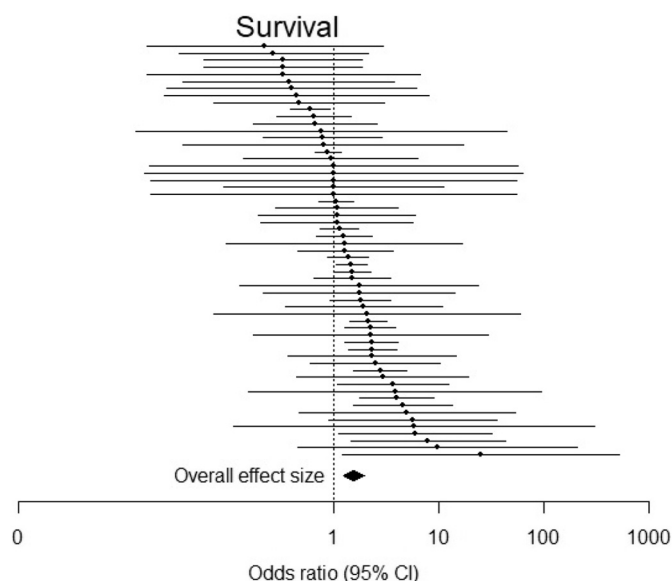


Fig. 1. Forest plot of all effects ($n = 59$) from 36 studies included in meta-analysis of survival for conditioned and unconditioned translocated animals. Points represent odds ratios and lines are 95% confidence intervals (CI) for study-level effects. The diamond represents the overall estimated odds ratio and 95% CI showing the odds of survival was higher for conditioned animals on average.

3.3. Movement

Distributions of effects for movement based on conditioning tactic, translocation type, age class, and taxonomic group are shown in Table 2. Soft release had the most effects, followed by environmental enrichment, and there was only one effect for antipredator training. There were more effects for captive releases than wild-to-wild translocations. There were slightly more effects for adults than juveniles, and only a few effects were from multiple age classes. The largest number of effects was for reptiles, followed by mammals, then fish, and there was only one effect for birds.

There was a small-to-moderate overall negative effect of conditioning on movement (Hedge's $g = -0.34$, 95% CI: $-0.63, -0.06$), indicating conditioned animals moved less than unconditioned ones on average post-release (Fig. 2). We detected moderate heterogeneity between effects ($I^2 = 51.56\%$). Soft released animals had reduced movement compared to hard released animals (Hedge's $g = -0.51$, 95% CI: -0.84 to -0.19). Compared to unenriched animals, environmental enrichment appeared to have minimal effects on reducing movement (Hedge's $g = 0.12$, 95% CI: -0.37 to 0.61). Conditioning reduced movement for wild-to-wild translocated animals compared to unconditioned ones (Hedge's $g = -0.60$, 95% CI: -1.17 to -0.04), but conditioning had minimal overall effects for animals released from captivity (Hedge's $g = -0.10$, 95% CI: -0.44 to 0.24). Conditioning reduced movement for adults (Hedge's $g = -0.48$, 95% CI: -0.87 to -0.08) but not juveniles (Hedge's $g = -0.01$, 95% CI: -0.27 to 0.24). Movement of mammals was reduced if they were conditioned (Hedge's $g = -0.35$, 95% CI: -0.69 to 0.00). There was more uncertainty regarding the effects of conditioning on movement for reptiles (Hedge's $g = -0.32$, 95% CI: -0.82 to 0.17). We found minor evidence of publication bias from funnel plots, but the Egger's regression test was non-significant ($z = -0.37$, $P = 0.71$). We found some potential for publication bias from the trim-and-fill analysis, which estimated four effects were missing (Fig. A5).

3.4. Site fidelity

Soft release was the only conditioning tactic used to increase site

Table 2

Contingency tables showing the distribution of effects for movement of conditioned and unconditioned translocated animals based on age class, taxonomic group, conditioning tactic, and translocation type. NA refers to nonsensical effects (e.g., wild-to-wild translocations would not use environmental enrichment since this conditioning type occurs in captivity).

Age class	Taxa			
	Actinopterygii	Aves	Mammalia	Reptilia
Adult	0	1	10	11
Juvenile	4	0	0	12
Multiple	0	0	1	2

Conditioning tactic	Age class		
	Adult	Juvenile	Multiple
Soft release	18	6	3
Environmental enrichment	3	10	0

Conditioning tactic	Taxa			
	Actinopterygii	Aves	Mammalia	Reptilia
Soft release	0	1	10	16
Environmental enrichment	4	0	0	9

Conditioning tactic	Translocation type	
	Captive release	Wild-to-wild
Soft release	11	16
Environmental enrichment	13	NA

Translocation type	Age class		
	Adult	Juvenile	Multiple
Captive release	8	16	0
Wild-to-wild	14	0	3

Translocation type	Taxa			
	Actinopterygii	Aves	Mammalia	Reptilia
Captive release	4	1	3	16
Wild-to-wild	0	0	8	9

fidelity of translocated animals. There were slightly more effects for wild-to-wild translocations than captive releases (Table 3). Most effects were from adults, two were for multiple age classes, and there was only one effect for juveniles. The largest number of effects was for reptiles, followed by mammals. There was only one effect for birds and none for fish.

Soft released animals were 3.20 (95% CI: 1.23 to 8.34) times more likely to show site fidelity than hard released animals (Fig. 3). We detected no evidence of heterogeneity in the data ($I^2 = 0.00\%$). Wild-to-wild translocated animals were 3.50 (95% CI: 1.48 to 8.30) times more likely to show site fidelity when soft released than if hard released, and this was the only subgroup comparison we were able to make for the site fidelity variable. We found minor evidence of publication bias from the funnel plot, likely due to the low number of effects analyzed, but the Egger's regression test was not significant ($z = 1.37$, $P = 0.17$). The trim-and-fill analysis estimated one effect was missing (Fig. A6).

4. Discussion

Outcomes of wildlife translocations are variable but often result in

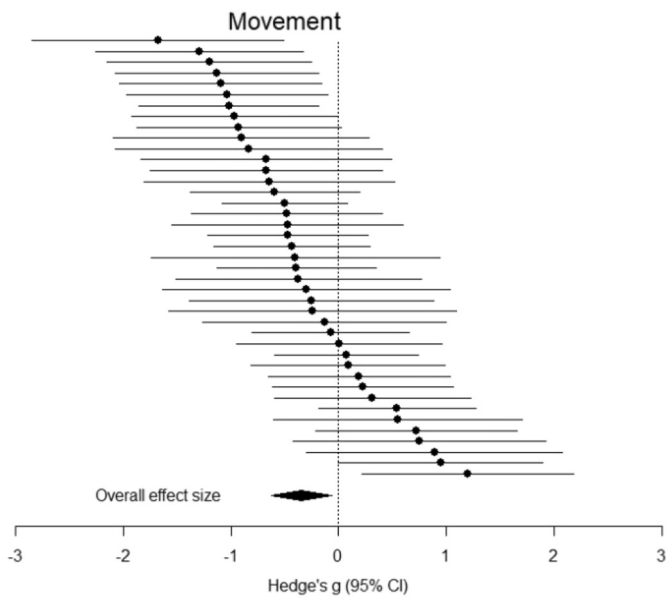


Fig. 2. Forest plot of all effects ($n = 41$) from 18 studies included in meta-analysis of movement for conditioned and unconditioned translocated animals. Points represent standardized mean difference (Hedge's g) and lines are 95% confidence intervals (CI) for study-level effects. The diamond represents the overall estimated standardized mean difference and 95% CI showing conditioned animals moved less than unconditioned ones on average.

Table 3
Contingency tables showing the distribution of effects for site fidelity of conditioned and unconditioned translocated animals based on age class, taxonomic group, and translocation type. Only soft release was used as a conditioning tactic.

Age class	Taxa		
	Aves	Mammalia	Reptilia
Adult	1	2	2
Juvenile	0	0	1
Multiple	0	0	2

Translocation type	Age class		
	Adult	Juvenile	Multiple
Captive release	2	1	0
Wild-to-wild	3	0	2

Translocation type	Taxa		
	Aves	Mammalia	Reptilia
Captive release	1	1	1
Wild-to-wild	0	1	4

failure (Fischer and Lindenmayer, 2000; Germano and Bishop, 2009). To date, several reviews have espoused the benefits of conditioning to improve conservation translocations (Fischer and Lindenmayer, 2000; Swaisgood, 2010; Reading et al., 2013), and some programs have seen high success, such as for recovery of black-footed ferrets (Dobson and Lyles, 2000) and whooping cranes (*Grus americana*) (Urbanek et al., 2010). However, a meta-analytical assessment to evaluate the efficacy of various conditioning tactics had not yet been conducted, likely due to the complicated nature of quantitatively synthesizing such a diverse literature. Put simply, the manner in which translocation programs implement conditioning varies widely between studies, making comparisons difficult. Despite these inherent difficulties, we believe there is

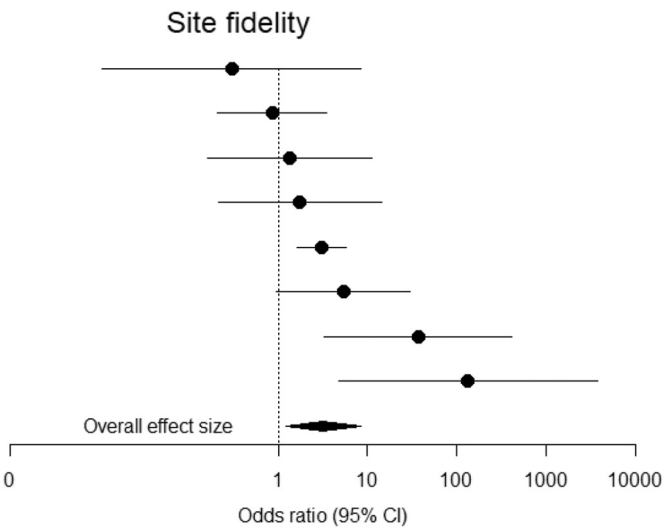


Fig. 3. Forest plot of all effects ($n = 8$) from six studies included in meta-analysis of site fidelity for soft released relative to hard released translocated animals. Points represent odds ratios (OR) and lines are 95% confidence intervals (CI) for study-level effects. The diamond represents the overall estimated OR and 95% CI showing the odds of exhibiting site fidelity was higher for soft released animals on average.

value in making broad comparisons between translocations using conditioning tactics and those that do not. Our results provide evidence that behavioral conditioning is on average beneficial for multiple benchmarks used to gauge translocation success. Although more demonstration and validation is necessary before wide-spread adoption can be advocated, our results indicate many translocation programs would benefit from the use of one or more conditioning tactics and could help guide practitioners in choosing the most appropriate tactic for their situation.

It is encouraging our analyses revealed both “animal-focused” (antipredator training and enrichment) and “environment-focused” (soft release) conditioning tactics (Batson et al., 2015) broadly led to higher survival for translocated animals. These effects were most pronounced when practitioners released juveniles from captivity. Naïve captive-reared animals might fail to develop essential learned behaviors that could enhance survival probability once released (Einum and Fleming, 2001; Mathews et al., 2005; Reading et al., 2013). Efforts expended in promoting natural behaviors appears to have positive effects on survival and is an area deserving of further attention. In contrast, conditioning had minimal effects on survival for wild-to-wild translocated animals, which commonly involve adults. Wild-to-wild translocations are often more successful than those releasing captive animals (Griffith et al., 1989; Fischer and Lindenmayer, 2000). Captive animals thus likely receive the greatest benefits from conditioning whereas practitioners translocating wild animals might experience less payoff. Additionally, young animals might exhibit greater behavioral plasticity than adults and thus be more likely to respond and change their behaviors in response to conditioning stimuli, whereas adults (particularly wild adults) might have more fixed behaviors. In short, practitioners could have the ability to condition beneficial behaviors for captive-reared juveniles whereas translocation of adults should focus on preventing degradation of behavior if animals are held in captivity for long periods (DeGregorio et al., 2017).

Translocated animals often make large, erratic movements as they orient to their new surroundings or attempt to home to their capture locations (Swaisgood, 2010). Our results support the idea that soft release via temporary penning is effective for confining wild adults until they develop some affinity for the release site and thus are less likely to make large movements upon release. We did not find consistent effects of soft release on the movements of captive or juvenile animals. This

approach might not have as much impact on these individuals because they generally do not have established home ranges like many wild adults do and are thus likely not attempting to home. Soft release also had the desired effect of increasing site fidelity. The influence of conditioning on site fidelity had the lowest number of effects across our datasets, so this is a focal area that deserves further investigation.

Our results also suggest the effectiveness of conditioning varies by taxon. We found the most pronounced effects of conditioning enhancing survival in fish, with somewhat less overall survival benefits observed for reptiles and mammals. We suggest the narrower confidence intervals around observed effect sizes for fish is a result of greater sample sizes. In our datasets, studies translocating birds, mammals, or reptiles generally released tens of animals, whereas most fish studies released hundreds or thousands of individuals. Translocating large numbers of animals has been associated with success (Wolf et al., 1998; Fischer and Lindenmayer, 2000), so releasing large groups of conditioned fish could have greater relative advantages compared to other vertebrate taxa. Conditioning had the most variable effects on survival for birds, demonstrating that avian translocation programs broadly need refinements to conditioning protocols. There have been several successes involving conditioning of birds, as has been demonstrated for Andean condors (*Vultur gryphus*) (Wallace and Temple, 1987), California condors (*Gymnogyps californianus*) (Meretsky et al., 2000), whooping and sandhill cranes (*Antigone canadensis*) (Duff et al., 2001; Urbanek et al., 2010), and Puerto Rican parrots (*Amazona vittata*) (White et al., 2005). These programs frequently rely upon time- and labor-intensive protocols, but more passive conditioning is unlikely to be successful for many bird translocations. Conditioning was effective for reducing movement of mammals but less so for reptiles. Limited evidence suggests lengthier conditioning programs might be most beneficial for reptiles. For example, soft release can be effective at reducing movement if reptiles are held in pens for several months to upwards of a year or more (Tuberville et al., 2005; Knox and Monks, 2014; Knox et al., 2017).

The limited number of studies we were able to include in our analyses compared to myriad conducted translocations raises two important points: First, our findings demonstrate the broad benefits of behavioral conditioning, so more research using these and other tactics (sensu Batson et al., 2015) is needed. Translocations are most frequently conducted using imperiled or game species, but future studies using common species would allow practitioners to experimentally assess the effects of these tactics in a scientifically rigorous manner, which could aid in refining our ability to successfully implement them. Second, several studies did experimentally implement the conditioning tactics we evaluated but reported insufficient information for inclusion in meta-analysis. We urge authors to report necessary information so analyses such as ours can be updated as more studies are conducted (Moseby et al., 2014; Nakagawa et al., 2017). Additionally, although translocations are diverse in their design and execution, adhering to existing frameworks such as the Translocation Tactics Classification System could facilitate standardization of tactics and thus data collation in future meta-analyses (Batson et al., 2015).

We identified several gaps in the literature that limit our ability to broadly evaluate the effects of behavioral conditioning on translocation success. We are unaware of or were unable to include experimental studies that used antipredator training for reptiles (antipredator training experiments were generally scant), translocations of conditioned adult fish, wild-to-wild translocations of conditioned fish, or wild-to-wild translocations comparing solely conditioned and unconditioned juveniles. Conditioning experiments with amphibians and invertebrates are also deficient despite potential for propagation of large release cohorts, similar to fish. We also found geographic gaps in the literature, with very few studies reported from Asia or South America. We do not suspect this biased the generality of our findings, but practitioners conducting translocations in these areas might consider increased adoption of conditioning experiments. Finally, based on

available literature we were limited to using short-term success measures in our analyses. Post-release monitoring lasted 157 days on average and ranged from 10 to 825 days across studies. Establishment and persistence of a translocated population without the need for continued human support is the ultimate goal of translocations (Seddon, 1999), so more long-term studies on the effects of conditioning on translocation success are needed.

The potential conservation value of wildlife translocations is great, but they can be risky and high-cost endeavors (Kleiman, 1989; Fischer and Lindenmayer, 2000). Lengthy and labor-intensive conditioning programs can also be very costly yet provide minimal benefits, as was found for initial pre-release conditioning efforts for reintroduction of golden lion tamarins (*Leontopithecus rosalia*) (Kleiman et al., 1991). Thus, deciding whether to incorporate a conditioning component could be an important consideration when designing translocation programs. Our meta-analyses suggest conditioning animals with any of the tactics we investigated has overall positive effects on survival, particularly for captive juveniles. Soft release might be the most effective tactic for reducing movement for wild-to-wild translocated adults. Soft release also appears useful for increasing site fidelity, but more experiments are needed to corroborate the limited number of studies we analyzed. Although environmental enrichment had minimal effects on movement, coupling enrichment with soft release could be an effective tactic for reducing movement and improving survival of animals released from captivity. Conditioning fish, reptiles, and mammals is generally effective, but behavioral conditioning associated with avian translocations needs refinement. Although our results show broad positive effects of behavioral conditioning on translocations, we emphasize previous authors in that a “one size fits all” approach does not apply for translocations given their system-specific attributes (Moseby et al., 2014). Nevertheless, we encourage continued conditioning experiments and adaptive management (Armstrong et al., 2007) to further illuminate how improved tactics could lead to even greater success in future efforts.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocon.2019.05.054>.

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